Subfield and torus geometries

Human Compression project

We realize every admissible parameter tuple in Constructions 8 and 9 of the fixed Suzuki source. A valuation calculation gives all intersections, including the cases in which a selected prime divides the base torus order. Descent and cyclic coset coordinates prove flag transitivity; the same intersections give thickness, residual connectedness and residual weak primitivity.

1 Source parameters and the base field

Write the field exponent as n, so the ambient group is $G = \mathrm{Sz}(2^n)$. The source chooses $k = \alpha - 1$ pairwise nonisomorphic maximal simple subfield groups $\mathrm{Sz}(2^{n/a_i})$, where the relative degrees a_i are distinct odd primes. Set

$$e = n/\prod_i a_i, \qquad t = 2^e.$$

The upper bound $\alpha \le \sum_j e_j = \Omega(n)$ in both constructions implies $\Omega(e) = \Omega(n) - k \ge 1$. Since n is odd, $e \ge 3$ **and** $t \ge 8$. Thus neither construction includes a nonsimple base $\mathrm{Sz}(2)$. The notation Ω here counts prime factors with multiplicity. Some a_i may also divide e; the proof allows that.

Construction 8 chooses distinct primes

$$\ell_i \mid \frac{t^{a_i} - 1}{t - 1}$$

and adds the dihedral subgroup of order

$$2(t-1)\prod_i \ell_i.$$

Construction 9 chooses a base nonsplit torus order $\gamma = t + \sqrt{2t} + 1$ or $t - \sqrt{2t} + 1$, and distinct primes ℓ_i such that $\gamma\ell_i$ divides an appropriate nonsplit torus order of $\mathrm{Sz}(t^{a_i})$. It also requires $\gamma \prod_i \ell_i$ to divide one of the two ambient nonsplit torus orders. Its added subgroup has order

$$4\gamma \prod_i \ell_i$$

and is cyclic by C_4.

The source does **not** require $\ell_i \nmid \gamma$. This note covers those cases in which a selected prime divides the base order explicitly. The arbitrary-rank existence theorem in the companion descent paper uses the extra restriction $a_i \nmid \tau_e$ only to prove an unconditional supply of parameters; that restriction is not needed for the present realization theorem for every admissible parameter tuple.

2 Both nonsplit branches have compatible binary models

The companion descent paper constructs a torus $\mathbf{T}$, stable under the exceptional endomorphism ρ, with

$$|\mathbf{T}^{\rho^d}| = \tau_d = 2^d + 1 - \big((1+i)^d + (1-i)^d\big)$$

for every odd d. To obtain the other branch, retain its binary order-four matrix r and antidiagonal w, and set

$$c = r^2 w.$$

The retained exact seed check gives

$$|c| = 5, \qquad \rho(c) = c, \qquad C_{\mathrm{Sz}(2)}(c) = \langle c \rangle.$$

It also verifies that r conjugates c to c^2 or c^3, so normalizes its centralizer. As the multiplicative order of 2 modulo 5 is four, c has four distinct eigenlines cyclically permuted by binary Frobenius. Therefore $\mathbf{T}' = C_{\mathrm{Sp}_4}(c)$ is another maximal torus with the same four-cycle Frobenius calculation as in the companion descent paper. It is ρ-stable because $\rho(c) = c$.

The character-lattice endomorphism has determinant two and trace ± 2; its five ρ-fixed points select trace **minus two**. For odd d, this gives

$$|(\mathbf{T}')^{\rho^d}| = \tau'_d = 2^d + 1 + \big((1+i)^d + (1-i)^d\big).$$

Both finite families are cyclic. Each satisfies

$$T_a \cap T_b = T_{\gcd(a,b)}, \qquad \gcd(|T_a|, |T_b|) = |T_{\gcd(a,b)}|. \tag{1}$$

The binary r supplies the order-four complement in every odd extension.

For any fixed odd base exponent e, these two branches give precisely the two possible base torus orders γ. Choose the branch whose order at e is γ, and write its order at d as h_d. Since $T_e \le T_{ea_i}$, $\gamma \mid h_{ea_i}$. The other order at ea_i is coprime to h_{ea_i}. Because $\gamma > 1$, it cannot also be divisible by γ. Consequently each source condition $\gamma\ell_i$ dividing a nonsplit torus order automatically selects this same compatible branch. The same argument identifies the prescribed ambient branch. Thus all the source's choices can be made simultaneously inside one ρ-stable torus.

For Construction 8 use the standard split tori

$$T_d = \{h(\lambda) : \lambda \in \mathbb{F}_{2^d}^{\times}\}, \qquad h_d = 2^d - 1.$$

They satisfy the same intersection and gcd identities (1). Their common complement is the binary w, of order two, acting by inversion. In the companion descent paper's standard matrix convention, the diagonal subgroups are compatible under every odd subfield embedding: when $d \mid n$ and n/d is odd, $(n+1)/2 \equiv (d+1)/2 \pmod{d}$, so the ambient exponent θ_n restricts to θ_d on $\mathbb{F}_{2^d}$.

3 One intersection calculation covers both constructions

Write $c_0 = 2$ for the split case and $c_0 = 4$ for either nonsplit case. Set $\gamma = t - 1$ in Construction 8; in Construction 9 retain the chosen nonsplit base order. Thus $\gamma = h_e$ in every case. In the chosen ambient cyclic torus T_n, let

$$A = C_m, \qquad m = \gamma \prod_i \ell_i, \qquad K = A : C_{c_0}.$$

For Construction 9, $m \mid h_n$ is a source assumption. For Construction 8 it follows from the individual divisibilities: each $\gamma\ell_i$ divides $2^n - 1$, and, because the primes are distinct, their least common multiple is m. The common binary complement normalizes A.

Take the standard subfield parabolics $S_i = \mathrm{Sz}(2^{n/a_i})$. For $J \subseteq \{1, \ldots, k\}$, put

$$n_J = e \prod_{i \notin J} a_i, \qquad m_J = \gamma \prod_{i \notin J} \ell_i.$$

Then

$$\bigcap_{j \in J} S_j = \mathrm{Sz}(2^{n_J}), \qquad K \cap \bigcap_{j \in J} S_j = C_{m_J} : C_{c_0}. \tag{2}$$

Here is the valuation argument that handles $\ell_i \mid \gamma$. For the prime ℓ_i, the exponent in m is $v_{\ell_i}(\gamma) + 1$. If $i \notin J$, $ea_i \mid n_J$, so $\gamma\ell_i \mid h_{n_J}$; the extra copy is retained. If $i \in J$, then

$$\gcd(ea_i, n_J) = e, \qquad \gcd(h_{ea_i}, h_{n_J}) = \gamma.$$

Since $\gamma\ell_i \mid h_{ea_i}$, this forces $v_{\ell_i}(h_{n_J}) = v_{\ell_i}(\gamma)$: exactly the extra copy is removed. Primes not selected among the ℓ_i occur in m only through γ, which is always retained. Thus $\gcd(m, h_{n_J}) = m_J$. The binary complement belongs to every standard subfield group, proving the full cyclic-by-complement equality in (2).

This proof allows $a_i \mid e$: factoring e from both exponents still gives $\gcd(ea_i, n_J) = e$, since the remaining relative primes are distinct from a_i.

4 Cyclic coset coordinates

Let $H_i = K \cap S_i = C_{m/\ell_i} : C_{c_0}$ and $A_i = A \cap H_i = C_{m/\ell_i}$. Every left coset of H_i in K has a representative in A, because the complement is contained in H_i, and

$$A/A_i \longrightarrow K/H_i, \qquad aA_i \longmapsto aH_i$$

is a bijection of sets.

The cyclic group A maps onto $\prod_i A/A_i$: these are cyclic quotients of the pairwise distinct prime orders ℓ_i, so this is the ordinary Chinese remainder theorem. Its kernel has order γ. This remains true when some selected primes divide γ. Hence the set map

$$K \longrightarrow \prod_i K/H_i$$

is surjective. No H_i is asserted to be normal in K. Only the subgroups A_i of the abelian A are used as group-quotient kernels.

The standard subfield types satisfy Theorem C of the companion descent paper. Its Lemma D now gives a common representative in every flag of the enlarged coset system. The system is a geometry, and G is flag-transitive.

5 RWPRI, thickness, and residual connectedness

The all-subset proof in the companion descent paper applies with (2). Before selecting K, each unused field type makes a maximal simple-subfield step; their terminal group is $\mathrm{Sz}(2^e)$, with $e \geq 3$. Selecting K there gives its full split or nonsplit torus normalizer, a maximal subgroup. After selecting K, an unused field type removes precisely one factor ℓ_i, giving prime subgroup index. These are the source's RWPRI conditions.

Panel sizes are the odd primes ℓ_i, and the index of the full base torus normalizer in the simple group $\mathrm{Sz}(2^e)$. The latter is greater than two (an index-two subgroup of a simple group would be normal). Thus the geometry is thick.

For residual connectedness, standard residues are enough because flag transitivity is proved. With K already selected, two unused field types give cyclic parts of orders m_J/ℓ_i and m_J/ℓ_j. Their least common multiple is m_J, so they generate the current cyclic-by-complement subgroup. Without K, two remaining distinct maximal simple subfields generate the current Suzuki group. If only one field type and K remain, the simple base subfield is maximal, and (2) shows that the other marked subgroup is not contained in it. They therefore generate as well. This proves connectedness of every residue of rank at least two, including the whole geometry.

We have thus rederived the complete parameterized **existence assertions** of Constructions 8 and 9, and proved flag transitivity and residual connectedness for these canonical realizations. Their inclusion in an exhaustive classification, the classification of all embeddings with those isomorphism types, and geometric intersection-property axioms are separate questions.